Complexes of stable ∞-categories

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Plan
1) Complexes, cubes and totalization
2) Normal crossing divisors
3) Fukaya-Seidel complexes

1) Rules of categorification

abelian group A	stable ∞-category $\mathcal{A}$
element $x \in A$	object $X \in \mathcal{A}$
difference $x - y$	require a map $f: x \to y$
	$\mathrm{fib}(x \xrightarrow{f} y) = \mathrm{cone}(f)[-1]$
$C = A \oplus B$ internal direct sum $A, B \subset C$	$\mathcal{C}$ has a semiorthogonal decomposition $(\mathcal{A}, \mathcal{B})$ (e.g. a recollement)

K_0

K_0

$A \oplus B$ external direct sum

$\cong A \times B$ product

require functor $F: \mathcal{A} \to \mathcal{B}$

lax limit

$\mathrm{lax\,lim}(\mathcal{A} \xrightarrow{F} \mathcal{B}) = \mathcal{A} \overset{\rightarrow}{\times}_F \mathcal{B}$

$$\begin{tikzcd} & \Rightarrow & \\ \mathcal{A} \arrow[rr, "F"] & & \mathcal{B} \end{tikzcd}$$

lax limit in the $(\infty,2)$-category of stable ∞-categories St

$\mathcal{A} \overset{\rightarrow}{\times}_F \mathcal{B}$ has objects (a,b,α) w/

$a \in \mathcal{A}$, $b \in \mathcal{B}$, $\alpha: F(a) \to b$ in $\mathcal{B}$.

It has a semiorthogonal decomposition w/ gluing functor F.

w/ $\mathcal{A} = \{(a,0,0)\} \subset \mathcal{A} \overset{\rightarrow}{\times}_F \mathcal{B}$

$\mathcal{B} = \{(0,b,0)\} \subset$

coproduct

$\mathrm{laxcolim} = \mathcal{A} \overset{\rightarrow}{\amalg}_F \mathcal{B}$

$$\begin{tikzcd} & \Leftarrow & \\ \mathcal{A} \arrow[ur] \arrow[rr, "F"'] & & \mathcal{B} \arrow[ul] \end{tikzcd}$$

Classically:

Ab is an additive category if

1) admits finite products

2) enriched in abelian groups

Then: finite products and coproducts coincide.

$\rightsquigarrow$ direct sum $\oplus$

categorified

St is a lax additive $(\infty,2)$-cat, meaning

1) finite lax limits

2) enriched in stable ∞-cats

Then (CDW)

finite lax limits coincide w/ finite lax colimits

$\rightsquigarrow$ justifies that $\mathcal{A} \overset{\rightarrow}{\oplus}_F \mathcal{B} := \mathcal{A} \overset{\rightarrow}{\times}_F \mathcal{B} \simeq \mathcal{A} \overset{\rightarrow}{\amalg}_F \mathcal{B}$

categorifies $\mathcal{A} \oplus \mathcal{B}$.

Framework of lax additive $(\infty,2)$-categories allows us to categorify additive aspects of homological algebra.

Chain complexes are an additive aspect.

<u>Def</u>: A categorical complex is a functor $\mathcal{A}_\bullet : \mathbb{Z}^{op} \to \mathrm{St}$, i.e.

$$\cdots \longrightarrow \mathcal{A}_{i+2} \xrightarrow{d} \mathcal{A}_{i+1} \xrightarrow{d} \mathcal{A}_i \to \cdots,$$

satisfying that $d^2 \in \mathrm{St}(\mathcal{A}_{i+2}, \mathcal{A}_i)$ is a zero object. $\leftarrow$ stable ∞-cat of functors

Ex Let $F: \mathcal{A} \to \mathcal{B}$ be a functor.

i) $\cdots 0 \to \mathcal{A} \xrightarrow{F} \mathcal{B} \to 0 \to \cdots$ is a cat. cplx

ii) $\cdots 0 \to \underset{\mathrm{fib}(F)}{\underset{\shortparallel}{\mathrm{Ker}(F)}} \hookrightarrow \mathcal{A} \to \mathcal{B} \to \mathcal{B}/\mathcal{A} \to 0 \to \cdots$

is a cat cplx.

Def A categorical n-cube is a functor

$I^n \to \mathrm{St}$, where $I = \{1 \to 0\} \subset \mathbb{Z}^{op}$.

e.g. for $n=2$, a 2-cube is a commutative square.

Totalalization

for abelian groups, $n=2$

$$C_* = \begin{tikzcd} A \arrow[r, shift left, "\alpha"] \arrow[d, "\alpha'"'] & B \arrow[d, "\beta'"] \\ C \arrow[r, "\beta"'] & D \end{tikzcd} \qquad \mathrm{tot}(C_*) = \begin{tikzcd} A \arrow[d, "(-\alpha', \alpha)"] \\ C \oplus B \arrow[d, "(\beta, \beta')"] \\ D \end{tikzcd}$$

For categorical cubes, $n=2$

$$C_* = \begin{tikzcd} \mathcal{A} \arrow[r, "F"] \arrow[d, "F'"'] & \mathcal{B} \arrow[d, "G'"] \\ \mathcal{C} \arrow[r, "G"'] & \mathcal{D} \end{tikzcd}$$

product totalization:

$$\mathrm{tot}^{\times}(\mathcal{C}_*) = \begin{array}{c} \mathcal{A} \\ \downarrow \\ \mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{D}} \mathcal{B} \\ \downarrow \\ \mathcal{D} \end{array} \leftarrow \infty\text{-cat w/ objects } (c, b, \alpha)$$

where $c \in \mathcal{C}$, $b \in \mathcal{B}$

$\alpha : G(c) \to G'(c)$ in $\mathcal{D}$

(called the directed pullback)

Has a SOD w/ component $\mathcal{C}, \mathcal{D}$

and gluing functor $G^R G^!$.

$\uparrow$ right adjoint

Differential:

$a \in \mathcal{A}$ $\quad d(a) = (F^!(a), F(a), \alpha)$ w/

$\alpha : G F^!(a) \simeq G' F(a)$ in $\mathcal{D}$

$(c, b, \alpha) \in \mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{D}} \mathcal{B}$ $\quad d((c, b, \alpha)) = \mathrm{fib}(\alpha) = \mathrm{cone}(\alpha)[-1] \in \mathcal{D}$

$d^2 = 0$: cone of an equivalence vanishes.

coproduct totalization is dual:

$$\mathrm{tot}^{\amalg}(\mathcal{C}_*) = \begin{array}{c} \mathcal{A} \\ \downarrow \\ \mathcal{C} \overset{\rightarrow}{\amalg} \mathcal{B} \\ \downarrow \\ \mathcal{D} \end{array} \leftarrow \text{has semiorthogonal decomposition}$$

w/ gluing functor $F F^R$.

$\uparrow$ right adjoint

For higher cubes, define totalization recursively.

E.g. for $n=3$

$$\begin{tikzcd}
A \arrow[r] \arrow[d] \arrow[dr] & B_1 \arrow[d] \arrow[dr] & \\
B_2 \arrow[r] \arrow[d] & C_1 \arrow[d] & \\
B_3 \arrow[r] \arrow[d] & C_2 \arrow[r] \arrow[d] & D \\
C_3 \arrow[r] & D &
\end{tikzcd}
\xrightarrow{\mathrm{tot}^\times}
\begin{tikzcd}
A \arrow[r] \arrow[d] & B_1 \arrow[d] \\
B_3 \times_{C_3} B_2 \arrow[r] \arrow[d] & C_2 \times_D C_1 \arrow[d] \\
C_3 \arrow[r] & D
\end{tikzcd}$$

$$\xrightarrow{\mathrm{tot}^\times}
\begin{tikzcd}
A \arrow[d] \\
(B_3 \times_{C_3} B_2) \times_{C_2 \times_D C_1} B_1 \arrow[d] \\
C_3 \times_D (C_2 \times_D C_1) \arrow[d] \\
D
\end{tikzcd}$$

<u>Rem</u> i) $\mathrm{tot}^{\times}/\mathrm{tot}^{\amalg}$ in general depends on the chosen total order of the coordinates

E.g.

$$\mathrm{tot}^{\times}\begin{pmatrix} A \to B \\ \downarrow \quad \downarrow \\ C \to D \end{pmatrix} \neq \mathrm{tot}^{\times}\begin{pmatrix} A \to C \\ \downarrow \quad \downarrow \\ B \to D \end{pmatrix}$$

ii) Expectation: tot does not depend on the chosen partial totalizations at adjacent coordinates.

<u>Def</u>: 1) A 2-cube

$$C_{\times} = \begin{tikzcd} A \arrow[r, "F"] \arrow[d, "F'"] & B \arrow[d, "G'"] \\ C \arrow[r, "G"] & D \end{tikzcd}$$

is called horizontally right adjointable, if the canonical natural transformation (mate)

$$F'F^R \longrightarrow G^R G'$$

is a natural equivalence $\Longleftrightarrow$

$$\begin{tikzcd} A & B \arrow[l, "F^R"'] \arrow[d, "G'"] \\ C \arrow[u, "F'"] & D \arrow[l, "G^R"] \end{tikzcd}$$

commutes

2) An n-cube is called Beck-Chevalley, if each face is horizontally + vertically right adjointable.

<u>Thm</u> (CDW)

If $\mathcal{C}_*$ is a BC n-cube, then

$$\mathrm{tot}^{\times}(\mathcal{C}_*) \simeq \mathrm{tot}^{\amalg}(\mathcal{C}_*).$$

2) Normal crossing divisors

Have functor $\mathrm{IndCoh}(-)/\mathrm{QCoh}(-) : \mathrm{Sch}_k \to \mathrm{St}$

k field w/ $\mathrm{char}(k)=0$. ↑ schemes/k

Thus any cube of schemes gives rise to a categorical cube and thus a categorical cplx via totalization.

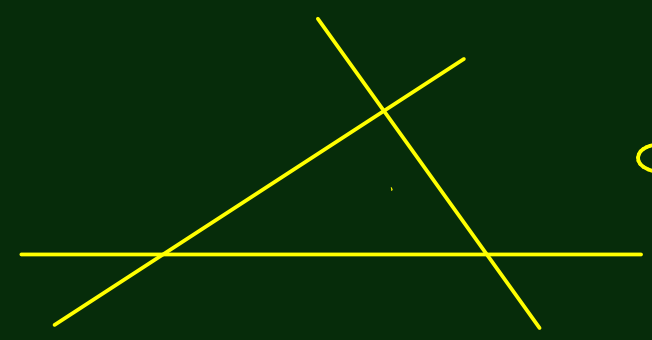
We consider a smooth proj variety Z and $D = \bigcup_{i=1}^{n} D_i \subset Z$ a normal crossing divisor,

i.e. $D_i \subset Z$ is a smooth divisor and repeated intersections of D_i's are smooth and of expected dimension

<u>Ex</u> $Z = \mathbb{P}^2$, D = union of three $\mathbb{P}^1$'s in general position

$D_i \cap D_j = pt \quad i \neq j$

$D_1 \cap D_2 \cap D_3 = \emptyset$.



$\subset \mathbb{P}^2$

We form a cube $D_* : I^n \to Sch_k$, e.g. for $n=2$

$$\begin{tikzcd} D_1 \cap D_2 \arrow[r] \arrow[d] & D_1 \arrow[d] \\ D_2 \arrow[r] & Z \end{tikzcd}$$

$\rightsquigarrow$ $Ind\,Coh(D_*)$ is a cat n-cube.

It is Beck-Chevalley by base-change.

$\mathcal{D}_* := tot(Ind\,Coh(D_*))$ is a cat. cplx, called the categorical intersection complex of D.

<u>In Ex</u>

$$\mathcal{D}_* = \quad 0 \to \mathcal{D}(pt)^{\times 3} \to \underbrace{\mathcal{D}(\mathbb{P}^1) \vec{\times} \mathcal{D}(\mathbb{P}^1) \vec{\times} \mathcal{D}(\mathbb{P}^1)}_{\mathcal{D}(pt) \vec{\times} \mathcal{D}(pt)} \to \mathcal{D}(\mathbb{P}^2) \to 0$$

Interesting properties of $\mathcal{D}_{\mathcal{A}}$:

1. $\mathcal{D}_{\mathcal{A}}$ is a spherical complex, meaning that the differentials are spherical functors

[Recall: Let $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ be an adjunction. F is called spherical if

* $T_{\mathcal{A}} := \mathrm{cof}(\mathrm{id}_{\mathcal{A}} \xrightarrow{\text{unit}} GF) \in \mathrm{Fun}(\mathcal{A}, \mathcal{A})$ is an equivalence

twist

* $T_{\mathcal{B}} := \mathrm{fib}(FG \xrightarrow{\text{counit}} \mathrm{id}_{\mathcal{B}}) \in \mathrm{Fun}(\mathcal{B}, \mathcal{B})$ is an equivalence

cotwist

This has to do w/ perverse sheaves:

<u>Thm (Gallingo–Granger–Masuoka '85)</u>

Equip $\mathbb{C}^n$ w/ stratification $\{0\}^n \subset \mathbb{C} \times \{0\}^{n-1} \subset \mathbb{C}^2 \times \{0\}^{n-2} \subset \cdots \subset \mathbb{C}^n$.

Then

$$\mathrm{Perv}(\mathbb{C}^n) \simeq \left\{ \text{diagrams of vector spaces } A_0 \underset{g_0}{\overset{f_0}{\rightleftarrows}} A_1 \underset{g}{\overset{f_1}{\rightleftarrows}} A_2 \cdots \underset{g_n}{\overset{f_n}{\rightleftarrows}} A_n \right\}$$

abelian cat of perverse sheaves

w/ $f^2 = 0$, $g^2 = 0$

$f_i g_i - \mathrm{id}_{A_i}$, $g_i f_i - \mathrm{id}_{A_{i+1}}$ are invertible

categorified by spherical complexes

2. If $D \subset Z$ is anticanonical, then $\mathcal{D}_*$ carries a Calabi-Yau structure. Calabi-Yau structures for complexes generalize Calabi-Yau structures for functors (due to Brav-Dyckerhoff).

These generalize usual Calabi-Yau structures

$$\left(R\mathrm{Hom}(X, Y) \simeq R\mathrm{Hom}(Y, X[n])^* \right)$$

3. Mirror symmetry for complexes w/ Fukaya-Seidel complexes.

3) Fukaya-Seidel complexes

Recall Seidel's construction (modulo technical details): Let X be an exact symplectic manifold.

Consider a $\underbrace{\text{Lefschetz fibration}}_{\text{complex analog of a Morse fibration. has constant fibres, except for finite set of mild singularities.}}$ $\pi : X \to \mathbb{C}$ w/ fibre X_1.

There is an A_∞-category $FS(\pi)$ w/ a sequence

$$\underset{\uparrow \text{ kernel of } \partial}{\mathrm{Fuk}(X)} \hookrightarrow FS(\pi) \xrightarrow{\partial} \underset{\uparrow \text{ Fukaya cat of cpt Lagrangians}}{\mathrm{Fuk}(X_1)}$$

$FS(\pi_1)$ has an exceptional collection, given by the <u>vanishing cycles</u>.

Lagrangian sphere inside of X_1, which contracts in a singularity of π.

Thus $FS(\pi_1)$ is much easier to handle than $Fuk(X)$.

<u>Remark</u>: ∂ is typically a spherical functor.

We iterate this construction, choosing a Lefschetz fibration

$$\pi_2 : X_1 \longrightarrow \mathbb{C} \quad \text{and so on.}$$

Get a categorical complex

$$\begin{tikzcd}
FS(X) \arrow[r, "d"] \arrow[dr, "\partial"'] & FS(X_1) \arrow[r, "d"] \arrow[dr, "\partial"'] & FS(X_2) \arrow[r, dashed] & \cdots \arrow[r] & Fuk(X_n) \\
& Fuk(X_1) \arrow[ur] & Fuk(X_2) \arrow[ur] & &
\end{tikzcd}$$

Fukaya-Seidel complex

Work in progress by Auroux,Ganatra, Hanlon, Jeffs produces categorical cubes from multi-potential Landau-Ginzburg models $(= X \to \mathbb{C}^n)$.

This should induce a collection of Lefschetz fibrations and totalizing their cube should produce the FS-complex.

Expectation FS-complexes can be mirror to categorical intersection complexes.

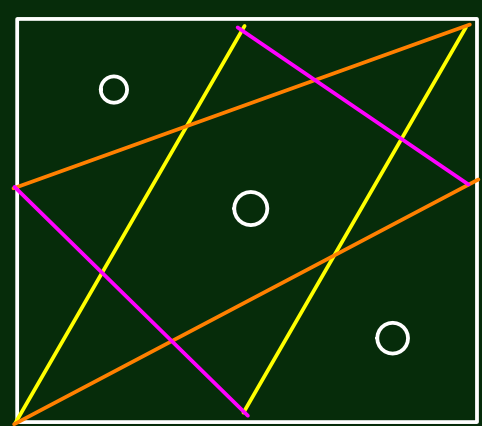
An example

$$X = (\mathbb{C}^*)^2 = \{xyz = 1\} \subset \mathbb{C}^3$$

$$\Big\downarrow \pi_1 = x+y+z$$

$$\mathbb{C}$$

Typical fibre is X_1 = elliptic curve w/ 3 punctures



3 vanishing cycles

$$FS(\pi_1) \simeq \mathcal{D}(pt) \vec{\times} \mathcal{D}(pt) \vec{\times} \mathcal{D}(pt) \overset{(!)}{\simeq} \mathcal{D}(\mathbb{P}^2)$$

Choose $\pi_2 = x - y : X_1 \to \mathbb{C}$, has typical fibre $X_2 = * \amalg * \amalg *$

$$FS(\pi_2) \underset{\uparrow}{\simeq} \mathcal{D}(pt)^{\vec{\times} 6} \overset{(!)}{\simeq} \underbrace{\mathcal{D}(\mathbb{P}^1)}_{\simeq \mathcal{D}(pt) \vec{\times} \mathcal{D}(pt)} \vec{\times} \mathcal{D}(\mathbb{P}^1) \vec{\times} \mathcal{D}(\mathbb{P}^1)$$

6 vanishing cycles

$$Fuk(X_2) \simeq \mathcal{D}(pt)^{\times 3}.$$

We recover the cat. intersection complex of 3 $\mathbb{P}^1$'s inside $\mathbb{P}^2$.